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Practical : WEEK 7

Task -1:

1. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Code:

```
UN PIC0 5.07 File: 70.c Modified
// Global segment
printf("Global address : %p\n", (void *)&global_var);

// Static segment
printf("Static global address : %p\n", (void *)&static_global_var);
printf("Static local address : %p\n", (void *)&static_var);

// Heap segment
printf("Heap address : %p\n", (void *)&heap_var);

// Stack segment
printf("Stack address : %p\n", (void *)&stack_var);
free(heap_var);

return 0;
}
```

Output:

```
Last login: Thu Sep 24 13:31:29 on console
nrx@nallas-mac-Book-Air ~ % nano 70.c

nrx@nallas-mac-Book-Air ~ % gcc 70.c -o 70
nrx@nallas-mac-Book-Air ~ % ./70
Linux Process Address Space
-----
Code address      : 0x1024804f8
Global address    : 0x102480000
Static global address : 0x102480008
Static local address : 0x102480004
Heap address      : 0x10247f000
Stack address     : 0x10497f1f8
nrx@nallas-mac-Book-Air ~ %
```

2. Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings

Code:

```
UW PICO 5.09 File: 71.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;

int main()
{
    int local_var = 200;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 300;

    printf("Process ID      : %d\n", getpid());
    printf("Global variable  : %p\n", (void *)&global_var);
    printf("Stack variable   : %p\n", (void *)&local_var);
    printf("Heap variable    : %p\n", (void *)&heap_var);

    printf("\nProcess is running...\n");
    printf("Press Ctrl+C to terminate.\n");

    while (1)
    {
        sleep(10);
    }

    free(heap_var);

    return 0;
}
```

Output:

```
nrk@nallas-mac-Book-Air ~ % nano 71.c
nrk@nallas-mac-Book-Air ~ % gcc 71.c -o 71
nrk@nallas-mac-Book-Air ~ % ./71
Process ID      : 1578
Global variable : 0x100478000
Stack variable  : 0x16f98f1f8
Heap variable   : 0x100d0da40

Process is running...
Press Ctrl+C to terminate.
^C
nrk@nallas-mac-Book-Air ~ %
```